



SRI VASAVI ENGINEERING COLLEGE (Autonomous)

PEDATADEPALLI, TADEPALLIGUDEM-534 101

Department of
Computer Science & Engineering (Artificial Intelligence)
and
Artificial Intelligence & Machine Learning

Date: 02.08.2024

Minutes of the Fourth Board of Studies

The Fourth Meeting of BOS, B.Tech in CSE(AI) and B.Tech in AI&ML was held at 3.00pm through online mode on 22.07.2024(Monday) using the following link:

<https://us02web.zoom.us/j/83140751300>

The following members attended the meeting:

S.No.	Name of the Member	Designation	Role
1.	Dr. Loshma Guniseti	Professor, HOD-AIML, SVEC	Chairperson
2.	Dr. Dasari Haritha	Professor , UCEK, Kakinada	University Nominee
3.	Dr Nagesh Bhattu Sristy	Asst.Professor, Department of CSE, NIT- AP	Academic Expert
4.	Dr. K. Venkata Rao	Professor, Dept of CS&SE, Andhra University.	Academic Expert
5.	Sri Seshagiri Telkapalli	Enterprise Architect, TCS Hyderabad	Industry Expert
6.	Sri T Vinay Kumar	Director, XpertBridge, Hyderabad	Industry Expert
7.	Sri M Jnana Surya Prakasha Rao	Pragmasys Consulting LLP, Gurgaon	Alumni
8.	Dr. G. V. N. S. R. Ratnakara Rao	Principal	Invited Member
9.	Dr. Ch. Rambabu	Dean, SA	Invited Member
10.	Mr. M. Subba Rao	Sr. Asst. Professor	Member
11.	Mr. M. Yesu Sekharam	Assistant Professor	Member
12.	Mr. Shaik Moulali	Assistant Professor	Member
13.	Mrs. P. Ujwala Sai	Assistant Professor	Member
14.	Mrs. M. Kiranmai	Assistant Professor	Member



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Item No. 1: Welcome note by the Chairperson BOS.

The HOD extended a formal welcome and introduced the members.

Item No. 2: Progress Report of the Department

Item No. 3: Review of Course Structure from I to II Semesters of B.Tech. CSE(Artificial Intelligence) and Artificial Intelligence & Machine Learning Programmes under V23 Regulation. The details are given in **Annexure-I**.

Item No. 4: Approval of Proposed Course Structure and Syllabus for III and IV Semesters of B.Tech.(CSE(AI)) and B.Tech.(AI&ML) Programmes under V23 Regulation

Approved Course Structure and Syllabus for III and IV Semesters of B.Tech in CSE(AI) and B.Tech in AI&ML Programmes under V23 Regulation, along with the following suggested changes:

Sem	Course Code	Course Name	Suggestions	Inclusions / Modifications
IV	V23AI T02	Machine Learning	In Machine Learning Course it was suggested that: 1. Remove Models Based on Decision Trees in Unit III 2. In Unit III, add Regularization and Feature Reduction Strategies like PCA 3. Include Spectral Clustering Tutorial by Ulrike von Luxburg in Unit V 4. Add Reference book Pattern Recognition and Machine Learning by Christopher M. Bishop	In Machine Learning Course 1. Removed Models Based on Decision Trees in Unit III 2. In Unit III, added Regularization and Principal Component Analysis topics 3. Added Spectral Clustering Tutorial by Ulrike von Luxburg in Unit V 4. Added Reference book Pattern Recognition and Machine Learning by Christopher M. Bishop



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The Approved and Modified Course Structure and Syllabus is given in **Annexure-II** and the Syllabus of Courses approved by BOS, CSE is given in **Annexure III**.

Chairperson of BOS
(Dr.Loshma Guniseti)



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AIML Department Progress Report

Academic Year 2023-24

Student Progress and Achievements

1. Results

B.Tech CSE(AI)

Batch	Semester	Registered	Passed	Fail	Pass %
2023-27	I	132	89	43	67.42
	II	131	97	34	74.05
2022-26	III	142	121	21	85.21
	IV	141	112	29	79.43
2021-25	V	69	62	7	89.86
	VI	69	56	13	81.16

B.Tech AI &ML

Batch	Semester	Registered	Passed	Fail	Pass %
2023-27	I	129	84	45	65.12
	II	129	97	32	75.19
2022-26	III	143	116	27	81.12
	IV	143	108	35	75.52
2021-25	V	69	59	10	85.51
	VI	69	57	12	82.61



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Student Activities and Achievements

Activity	Count
Virtual Internships	24
MOOCS (NPTEL / Other Certifications courses)	123
Global Certifications	45
Journals Publications	45 students 09 papers
Student Placements	Two students got placement offer of Rs. 6 lakhs p.a. in Third Year



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Faculty Activities and Achievements

Activity	Count
FDP/ SDP/ Workshops/ Refresher Courses attended by Faculty	42
MOOCS (NPTEL / Other Certifications courses)	22
Qualified in NET	01
Qualified in R-CET	01
Journals Publications	10
Conferences Publications	08
Patents Applied	05
Faculty as Research Supervisor	01
Awards received	02



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**Department of
Computer Science Engineering (Artificial Intelligence)
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Department Activities in Academic Year 2023-24

Activity	Count
Faculty Development Programmes Organized	03
Workshops / Seminars Organized	06
Boot camps organized	01
Number of Guest Lectures /Alumni Connect	04
Hackathons organized	02
Institution Innovation Cell (IIC) Activities	03
Aikyam Department Association Activities	06
Maitri Social Service Activities	02

Annexure -I

First Year Course Structure

I – Semester							
S.No.	Course Code	Category	Name of the Course	L	T	P	C
1.	V23ENT01	BS&H	Communicative English	2	0	0	2
2.	V23PHT01	BS&H	Engineering Physics	3	0	0	3
3.	V23MAT01	BS&H	Linear Algebra & Calculus	3	0	0	3
4.	V23CMT01	Engineering Science	Basic Civil & Mechanical Engineering	3	0	0	3
5.	V23CST01	Engineering Science	Introduction to Programming	3	0	0	3
6.	V23ENL01	BS&H	Communicative English Lab	0	0	2	1
7.	V23PHL01	BS&H	Engineering Physics Lab	0	0	2	1
8.	V23MEL01	Engineering Science	Engineering Workshop	0	0	3	1.5
9.	V23CSL01	Engineering Science	Computer Programming Lab	0	0	3	1.5
10.	V23CSL02	Engineering Science	IT Workshop	0	0	2	1
11.	V23SPT02	BS&H	Health and wellness, Yoga and sports	-	-	1	0.5
Total				14	0	13	20.5

L-Lecture, T-Tutorial, P-Practical, C-Credits

II – Semester							
S.No.	Course Code	Category	Name of the Course	L	T	P	C
1.	V23CHT01	BS&H	Applied Chemistry	3	0	0	3
2.	V23MAT02	BS&H	Differential Equations & Vector Calculus	3	0	0	3
3.	V23EET01	Engineering Science	Basic Electrical and Electronics Engineering	3	0	0	3
4.	V23MET01	Engineering Science	Engineering Graphics	1	0	4	3
5.	V23CST02	Professional Core	Data Structures	3	0	0	3
6.	V23CHL01	BS&H	Applied Chemistry Lab	0	0	2	1
7.	V23EEL01	Engineering Science	Electrical and Electronics Engineering Workshop	0	0	3	1.5
8.	V23CSL03	Professional Core	Data Structures Lab	0	0	3	1.5
9.	V23SPT01	BS&H	NSS/NCC/Scouts & Guides/Community Service	-	-	1	0.5
Total				13	0	15	20.5

L-Lecture, T-Tutorial, P-Practical, C-Credits

Annexure-II

III – Semester							
S.No.	Course Code	Category	Course	L	T	P	C
1.	V23MAT05	BS&H	Discrete Mathematics & Graph Theory	3	0	0	3
2.	V23MBT53	BS&H	Universal human values – understanding harmony and Ethical human conduct	2	1	0	3
3.	V23AIT01	Engineering Science	Artificial Intelligence	3	0	0	3
4.	V23CST04	Professional Core	Advanced Data Structures & Algorithm Analysis	3	0	0	3
5.	V23CST05	Professional Core	Object Oriented Programming through Java	3	0	0	3
6.	V23CSL04	Professional Core	Advanced Data Structures and Algorithm Analysis Lab	0	0	3	1.5
7.	V23CSL05	Professional Core	Object Oriented Programming Through Java Lab	0	0	3	1.5
8.	V23CSSE01	Skill Enhancement Course	Python Programming Lab	0	1	2	2
9.	V23MET09	BS&H	Design Thinking & Innovation	1	0	2	2
Total				15	02	10	22

L-Lecture, T-Tutorial, P-Practical, C-Credits

IV – Semester							
S.No.	Course Code	Category	Course	L	T	P	C
1.	V23MAT07	Engineering Science/Basic Science	Probability & Statistics	3	0	0	3
2.	V23MAT09	Management Course - I	Optimization Techniques	2	0	0	2
3.	V23AIT02	Professional Core	Machine Learning	3	0	0	3
4.	V23CST03	Engineering Science	Digital Logic & Computer Organization	3	0	0	3
5.	V23CST07	Professional Core	Database Management Systems	3	0	0	3
6.	V23AIL01	Professional Core	AI & ML Lab	0	0	3	1.5
7.	V23CSL07	Professional Core	Database Management Systems Lab	0	0	3	1.5
8.	V23CSSE02	Skill Enhancement Course	Full Stack Development –I	0	1	2	2
9.	V23CEAC01	Audit Course	Environmental Science	2	0	0	-
Total				16	01	08	19
Mandatory Community Service Project (V23COSP01) Internship of 08 weeks duration during summer vacation							

L-Lecture, T-Tutorial, P-Practical, C-Credits

Semester	III	L	T	P	C	COURSE CODE
Regulation	V23	3	0	0	3	V23AIT01
Name of the Course	Artificial Intelligence					
Branch	Common to CSE(AI) and AI&ML					

Syllabus Details

Course Outcomes: After Successful completion of the Course, the student will be able to:

CO1: Discuss the foundational concepts of AI. **(K2)**

CO2: Demonstrate Search Strategies for problem solving. **(K3)**

CO3: Illustrate Knowledge Representation approaches. **(K3)**

CO4: Demonstrate Reasoning approaches. **(K3)**

CO5: Illustrate architecture and role of Expert Systems. **(K3)**

UNIT - I

Introduction: AI problems, foundation of AI and history of AI intelligent agents: Agents and Environments, the concept of rationality, the nature of environments, structure of agents, problem solving agents, problem formulation.

UNIT - II

Searching: Searching for solutions, uniformed search strategies – Breadth first search, depth first Search. Search with partial information (Heuristic search) Hill climbing, A*, AO* Algorithms, Problem reduction, Game Playing-Adversarial search, Games, mini-max algorithm, optimal decisions in multiplayer games, Problem in Game playing, Alpha-Beta pruning, Evaluation functions.

UNIT - III

Representation of Knowledge: Knowledge representation issues, predicate logic- logic programming, semantic nets- frames and inheritance, constraint propagation, representing knowledge using rules, rules based deduction systems. Reasoning under uncertainty, review of probability, Bayes' probabilistic interferences and Dempster Shafer theory.

UNIT - IV

Logic concepts: First order logic, Inference in First order logic, Propositional vs. First Order inference, Unification & Lifting, Forward chaining, Backward chaining, Resolution, Learning from observation Inductive learning, Decision trees, Explanation based learning, Statistical Learning methods, Reinforcement Learning.

UNIT - V

Expert Systems: Architecture of expert systems, Roles of expert systems – Knowledge Acquisition Meta knowledge Heuristics. Typical expert systems – MYCIN, DART, XCON: Expert systems shells.

Textbooks:

1. Artificial Intelligence – A Modern Approach, S. Russel and P. Norvig, Third Edition, Pearson Education.
2. Artificial Intelligence, Elaine Rich and Kevin Knight, Shivashankar B Nair, Tata Mc Graw Hill

Reference Books:

1. Computational Intelligence: a logical approach, David Poole, Alan Mackworth, Randy Goebel, Oxford University Press.
2. Artificial Intelligence: Structures and Strategies for complex problem solving, G.Luger, Fourth Edition, Pearson Education.
3. Artificial Intelligence: A new Synthesis, J. Nilsson, Elsevier Publishers.
4. Artificial Intelligence, Saroj Kaushik, Cengage Learning.

Online Learning Resources:

1. <https://ai.google/>
2. https://swayam.gov.in/nd1_noc19_me71/preview

Semester	IV	L	T	P	C	COURSE CODE
Regulation	V23	3	0	0	3	V23AIT02
Name of the Course	Machine Learning					
Branch	Common to CSE(AI) and AI&ML					

Syllabus Details

Course Outcomes: After Successful completion of the Course, the student will be able to:

CO1: Illustrate various learning techniques and concepts of model.(K2)

CO2: Demonstrate Nearest Neighbor-Based Models.(K3)

CO3: Illustrate Bayes Classifier.(K3)

CO4: Demonstrate the concepts of Linear Discriminants and Regression.(K3)

CO5: Illustrate various clustering approaches.(K3)

UNIT-I: Introduction to Machine Learning: Evolution of Machine Learning, Paradigms for ML, Learning by Rote, Learning by Induction, Reinforcement Learning, Types of Data, Matching, Stages in Machine Learning, Data Acquisition, Feature Engineering, Data Representation, Model Selection, Model Learning, Model Evaluation, Model Prediction, Search and Learning, Data Sets.

UNIT-II: Nearest Neighbor-Based Models: Introduction to Proximity Measures, Distance Measures, Non-Metric Similarity Functions, Proximity Between Binary Patterns, Different Classification Algorithms Based on the Distance Measures ,K-Nearest Neighbor Classifier, Radius Distance Nearest Neighbor Algorithm, KNN Regression, Performance of Classifiers, Performance of Regression Algorithms.

UNIT-III: The Bayes Classifier: Introduction to the Bayes Classifier, Bayes' Rule and Inference, The Bayes Classifier and its Optimality, Multi-Class Classification, Class Conditional Independence and Naive Bayes Classifier (NBC),

Regularization Impact of Regularization in ML, **Principal Component Analysis** Maximum variance formulation, Minimum-error formulation, Applications of PCA , PCA for high-dimensional data

UNIT-IV: Linear Discriminants for Machine Learning: Introduction to Linear Discriminants, Linear Discriminants for Classification, Perceptron Classifier, Perceptron Learning Algorithm, Support Vector Machines, Linearly Non-Separable Case, Non-linear SVM, Kernel Trick, Logistic Regression, Linear Regression, Multi-Layer Perceptrons (MLPs), Back propagation for Training an MLP.

UNIT-V: Clustering : Introduction to Clustering, Partitioning of Data, Matrix Factorization, Clustering of Patterns, Divisive Clustering, Agglomerative Clustering, Partitional Clustering, K-Means Clustering, Soft Partitioning, Soft Clustering, Fuzzy C-Means Clustering, Rough Clustering, Rough K-Means Clustering Algorithm, Expectation Maximization-Based Clustering, Spectral Clustering Tutorial.

Text Books:

1. Machine Learning Theory and Practice, M N Murthy, V S Ananthanarayana, Universities Press (India), 2024

Reference Books:

1. Machine Learning, Tom M. Mitchell, McGraw-Hill Publication, 2017
2. Machine Learning in Action, Peter Harrington, DreamTech
3. Introduction to Data Mining, Pang-Ning Tan, Michel Steinbach, Vipin Kumar, 7th Edition, 2019.
4. Pattern Recognition and Machine Learning, Christopher M. Bishop, Springer
5. Ulrike Von Luxburg, "A tutorial on spectral clustering", Statistics and computing 17(2007): 395-416.

Semester	IV	L	T	P	C	COURSE CODE
Regulation	V23	0	0	3	1.5	V23AIL01
Name of the Course	AI & ML LAB					
Branch	Common to CSE(AI) and AI&ML					

Syllabus Details

Course Outcomes: After Successful completion of the Course, the student will be able to:

CO1: Demonstrate the Pandas Library for data manipulation and visualization. **(K3)**

CO2: Demonstrate Searching techniques. **(K3)**

CO3: Illustrate the pre-processing approaches. **(K3)**

CO4: Demonstrate Classification approaches **(K3)**

Software Required for ML: Python / R / Weka

List of Experiments

- Pandas Library
 - Demonstrate a Python program to implement Pandas Series with labels.
 - Prepare a Pandas Series from a dictionary.
 - Prepare a Pandas Data Frame.
 - Illustrate a program which makes use of the following Pandas methods
 - describe ()
 - head ()
 - tail ()
 - info ()
- Pandas Library: Visualization
 - Demonstrate a program which use Pandas inbuilt visualization to plot following graphs:
 - Bar plots
 - Histograms
 - Line plots
 - Scatter plots
- Demonstrate a program to Implement Breadth First Search using Python.
- Demonstrate a program to implement Best First Search Algorithm
- Demonstrate a program to Implement Depth First Search using Python.
- Demonstrate a program to implement the Heuristic Search
- Demonstrate a Python program to implement A* and AO* algorithm. (Ex: find the shortest path)
- Apply the following Pre-processing techniques for a given dataset.
 - Attribute selection
 - Handling Missing Values
 - Discretization
 - Elimination of Outliers
- Apply KNN algorithm for classification and regression

10. Demonstrate decision tree algorithm for a classification problem and perform parameter tuning for better results
11. Apply Random Forest algorithm for classification and regression
12. Demonstrate Naïve Bayes Classification algorithm.
13. Apply Support Vector algorithm for classification
14. Illustrate the K-means algorithm and apply it to the data you selected. Evaluate performance by measuring the sum of the Euclidean distance of each example from its class center. Test the performance of the algorithm as a function of the parameters K.

Reference Books:

1. Machine Learning Theory and Practice, M N Murthy, V S Ananthanarayana, Universities Press (India), 2024
2. Machine Learning, Tom M. Mitchell, McGraw-Hill Publication, 2017
3. Machine Learning in Action, Peter Harrington, DreamTech
4. Introduction to Data Mining, Pang-Ning Tan, Michel Steinbach, Vipin Kumar, 7th Edition, 2019.
5. Pattern Recognition and Machine Learning, Christopher M. Bishop, Springer

Annexure-III

Semester	III	L	T	P	C	COURSE CODE
Regulation	V23	3	0	0	3	V23CST04
Name of the Course	Advanced Data Structures & Algorithm Analysis					
Branch	Common to CSE, CST, CSE(AI) , AI&ML and CSE(DS)					

Syllabus Details

Course Outcomes: After Successful completion of the Course, the student will be able to:

CO1: Demonstrate asymptotic notation and non-linear data structures like AVL and B-Trees.(K3)

CO2: Demonstrate graphs and divide and conquer technique.(K3)

CO3: Use greedy approach to determine various problems.(K3)

CO4: Apply Dynamic Programming and Backtracking techniques.(K3)

CO5: Solve different problems using Branch and Bound, NP Hard and NP Complete.(K3)

UNIT-I

Introduction to Algorithm Analysis, Space and Time Complexity analysis, Asymptotic Notations.

AVL Trees – Creation, Insertion, Deletion operations and Applications.

B-Trees– Creation, Insertion, Deletion operations and Applications.

UNIT-II

Heap Trees (Priority Queues) – Min and Max Heaps, Operations and Applications.

Graphs – Terminology, Representations, Basic Search and Traversals, Connected Components and Biconnected Components, applications

Divide and Conquer: The General Method, Quick Sort, Merge Sort, Strassen's matrix multiplication, Convex Hull.

UNIT-III

Greedy Method: General Method, Job Sequencing with deadlines, Knapsack Problem, Minimum cost spanning trees, Single Source Shortest Paths– Dijkstra's .

UNIT-IV

Dynamic Programming: General Method, All pairs shortest paths, Single Source Shortest Paths–General Weights (Bellman Ford Algorithm), Optimal Binary Search Trees, 0/1 Knapsack, String Editing, Travelling Salesperson problem.

Backtracking: General Method, 8-Queens Problem, Sum of Subsets problem, Graph Coloring, 0/1 Knapsack Problem.

UNIT-V

Branch and Bound: The General Method, 0/1 Knapsack Problem, Travelling Salesperson problem.

NP Hard and NP Complete Problems: Basic Concepts, Cook's theorem (without Proof).

NP Hard Graph Problems: Clique Decision Problem (CDP), Traveling Salesperson Decision Problem (TSP).

NP Hard Scheduling Problems: Job Shop Scheduling.

Textbooks:

1. Fundamentals of Data Structures Horowitz, Ellis; Sahni, Sartaj; Mehta, Dinesh, 2nd Edition Universities Press

Reference Books:

1. Data Structures and program design in C, Robert Kruse, Pearson Education Asia
2. An introduction to Data Structures with applications, Trembley & Sorenson, McGraw Hill
3. The Art of Computer Programming, Vol.1: Fundamental Algorithms, Donald E Knuth, Addison-Wesley, 1997.
4. Data Structures using C & C++: Langsam, Augenstein & Tanenbaum, Pearson, 1995
5. Algorithms + Data Structures & Programs, N. Wirth, PHI
6. Fundamentals of Data Structures in C++: Horowitz Sahni & Mehta, Galgotia Pub.
7. Data structures in Java, Thomas Standish, Pearson Education Asia

Online Learning Resources:

1. https://www.tutorialspoint.com/advanced_data_structures/index.asp
2. <http://peterindia.net/Algorithms.html>
3. Abdul Bari, Introduction to Algorithms (youtube.com)

Semester	III	L	T	P	C	COURSE CODE
Regulation	V23	3	0	0	3	V23CST05
Name of the Course	Object Oriented Programming Through Java					
Branch	Common to CSE, CST, CSE(AI) , AI&ML and CSE(DS)					

Syllabus Details

Course Outcomes: After Successful completion of the Course, the student will be able to:

CO1:Demonstrate the object-oriented programming principles with Java programming environment. **(K3)**

CO2:Demonstrate the concepts like classes, objects, argument passing mechanism, overloading and overriding. **(K3)**

CO3:Illustrate the concepts of arrays, inheritance and interfaces. **(K3)**

CO4: Demonstrate packages , java libraries, exception handling, java I/O and File concepts. **(K3)**

CO5:Illustrate the concepts of string handling, multithreading and Java FX GUI. **(K3)**

UNIT-I

Object Oriented Programming: Basic concepts, Principles, Program Structure in Java: Introduction, Writing Simple Java Programs, Elements or Tokens in Java Programs, Java Statements, Command Line Arguments, User Input to Programs, Escape Sequences Comments, Programming Style.

Data Types, Variables, and Operators: Introduction, Data Types in Java, Declaration of Variables, Data Types, Type Casting, Scope of Variable Identifier, Literal Constants, Symbolic Constants, Formatted Output with printf() Method, Static Variables and Methods, Attribute Final, Introduction to Operators, Precedence and Associativity of Operators, Assignment Operator (=), Basic Arithmetic Operators, Increment (++) and Decrement (- -) Operators, Ternary Operator, Relational Operators, Boolean Logical Operators, Bitwise Logical Operators.

Control Statements: Introduction, if Expression, Nested if Expressions, if-else Expressions, Ternary Operator?:, Switch Statement, Iteration Statements, while Expression, do-while Loop, for Loop, Nested for Loop, For-Each for Loop, Break Statement, Continue Statement.

UNIT-II

Classes and Objects: Introduction, Class Declaration and Modifiers, Class Members, Declaration of Class Objects, Assigning One Object to Another, Access Control for Class Members, Accessing Private Members of Class, Constructor Methods for Class, Overloaded Constructor Methods, Nested Classes, Final Class and Methods, Passing Arguments by Value and by Reference, Keyword this.

Methods: Introduction, Defining Methods, Overloaded Methods, Overloaded Constructor Methods, Class Objects as Parameters in Methods, Access Control, Recursive Methods, Nesting of Methods, Overriding Methods, Attributes Final and Static.

UNIT-III

Arrays: Introduction, Declaration and Initialization of Arrays, Storage of Array in Computer Memory, Accessing Elements of Arrays, Operations on Array Elements, Assigning Array to Another Array, Dynamic Change of Array Size, Sorting of Arrays, Search for Values in Arrays, Class Arrays, Two-dimensional Arrays, Arrays of Varying Lengths, Three- dimensional Arrays, Arrays as Vectors.

Inheritance: Introduction, Process of Inheritance, Types of Inheritances, Universal Super Class-Object Class, Inhibiting Inheritance of Class Using Final, Access Control and Inheritance, Multilevel Inheritance, Application of Keyword Super, Constructor Method and Inheritance, Method Overriding, Dynamic Method Dispatch, Abstract Classes, Interfaces and Inheritance.

Interfaces: Introduction, Declaration of Interface, Implementation of Interface, Multiple Interfaces, Nested Interfaces, Inheritance of Interfaces, Default Methods in Interfaces, Static Methods in Interface, Functional Interfaces, Annotations.

UNIT-IV

Packages and Java Library: Introduction, Defining Package, Importing Packages and Classes into Programs, Path and Class Path, Access Control, Packages in Java SE, Java.lang Package and its Classes, Class Object, Enumeration, class Math, Wrapper Classes, Auto- boxing and Auto-unboxing, Java util Classes and Interfaces, Formatter Class, Random Class, Time Package, Class Instant (java.time.Instant), Formatting for Date/Time in Java, Temporal Adjusters Class, Temporal Adjusters Class.

Exception Handling: Introduction, Hierarchy of Standard Exception Classes, Keywords throws and throw, try, catch, and finally Blocks, Multiple Catch Clauses, Class Throwable, Unchecked Exceptions, Checked Exceptions.

Java I/O and File: Java I/O API, standard I/O streams, types, Byte streams, Character streams, Scanner class, Files in Java(Text Book 2)

UNIT-V

String Handling in Java: Introduction, Interface Char Sequence, Class String, Methods for Extracting Characters from Strings, Comparison, Modifying, Searching; Class String Buffer.

Multithreaded Programming: Introduction, Need for Multiple Threads Multithreaded Programming for Multi-core Processor, Thread Class, Main Thread-Creation of New Threads, Thread States, Thread Priority-Synchronization, Deadlock and Race Situations, Inter-thread Communication - Suspending, Resuming, and Stopping of Threads.

Java FX GUI: Java FX Scene Builder, Java FX App Window Structure, displaying text and image, event handling, laying out nodes in scene graph, mouse events (Text Book 3)

Textbooks:

1. Java one step ahead, Anitha Seth, B.L.Juneja, Oxford.
2. Joy with Java, Fundamentals of Object Oriented Programming, DebasisSamanta, MonalisaSarma, Cambridge, 2023.
3. Java 9 for Programmers, Paul Deitel, Harvey Deitel, 4th Edition, Pearson.

Reference Books:

1. The complete Reference Java, 11th edition, Herbert Schildt, TMH
2. Introduction to Java programming, 7th Edition, Y Daniel Liang, Pearson

Online Learning Resources:

1. <https://nptel.ac.in/courses/106/105/106105191/>
2. https://infyspringboard.onwingspan.com/web/en/app/toc/lex_auth_012880464547618816347_shared/overview

Semester	III	L	T	P	C	COURSE CODE
Regulation	V23	0	0	3	1.5	V23CSL04
Name of the Course	Advanced Data Structures & Algorithm Analysis Lab					
Branch	Common to CSE, CST, CSE(AI) , AI&ML and CSE(DS)					

Syllabus Details

Course Outcomes: After Successful completion of the Course, the student will be able to:

CO1: Demonstrate programs on AVL Trees and Heap trees. **(K3)**

CO2: Develop programs on Sorting algorithms and Graph traversal algorithms. **(K3)**

CO3: Develop programs using Greedy and Dynamic programming technique. **(K3)**

CO4: Develop programs using Backtracking and branch & Bound technique. **(K3)**

Experiments covering the Topics:

- Operations on AVL trees, Heap Trees
- Graph Traversals
- Sorting techniques
- Minimum cost spanning trees
- Shortest path algorithms
- 0/1 Knapsack Problem
- Travelling Salesperson problem
- Optimal Binary Search Trees
- N-Queens Problem
- Job Sequencing

Sample Programs:

1. Construct an AVL tree for a given set of elements which are stored in a file. And implement insert and delete operation on the constructed tree. Write contents of tree into a new file using in-order.
2. Construct B-Tree an order of 5 with a set of 100 random elements stored in array. Implement searching, insertion and deletion operations.
3. Construct Min and Max Heap using arrays, delete any element and display the content of the Heap.
4. Demonstrate BFT and DFT for given graph, when graph is represented by
 - a) Adjacency Matrix
 - b)Adjacency Lists
5. Develop a program for finding the biconnected components in a given graph.
6. Implement Quick sort and Merge sort and observe the execution time for various input sizes(Average, Worst and Best cases).
7. Examine the performance of Single Source Shortest Paths using Greedy method when the graph is represented by adjacency matrix and adjacency lists.
8. Demonstrate Job Sequencing with deadlines using Greedy strategy.
9. Demonstrate a program to solve 0/1Knapsack problem Using Dynamic Programming.
10. Demonstrate N-Queens Problem Using Backtracking.
11. Use Backtracking strategy to solve 0/1 Knapsack problem.
12. Demonstrate Travelling Sales Person problem using Branch and Bound approach.

Reference Books:

1. Fundamentals of Data Structures, Horowitz Ellis, SahniSartaj, Mehta, Dinesh, 2ndEdition, Universities Press
2. Computer Algorithms/C++ Ellis Horowitz, SartajSahni, SanguthevarRajasekaran, 2ndEdition, University Press
3. Data Structures and program design in C, Robert Kruse, Pearson Education Asia
4. An introduction to Data Structures with applications, Trembley& Sorenson, McGraw Hill

Online Learning Resources:

1. <http://cse01-iiith.vlabs.ac.in/>
2. <http://peterindia.net/Algorithms.html>

Semester	III	L	T	P	C	COURSE CODE
Regulation	V23	0	0	3	1.5	V23CSL05
Name of the Course	Object Oriented Programming Through Java Lab					
Branch	Common to CSE, CST, CSE(AI) , AI&ML and CSE(DS)					

Syllabus Details

Course Outcomes: After Successful completion of the Course, the student will be able to:

CO1:Construct programs to handle classes and objects. (K3)

CO2: Develop programs that incorporate inheritance and interfaces. (K3)

CO3:Construct programs on exception handling and File I/O. (K3)

CO4:Develop programs using multithreading and Java FX. (K3)

Experiments covering the Topics:

- Object Oriented Programming fundamentals- data types, control structures
- Classes, methods, objects, Inheritance, polymorphism,
- Exception handling, Threads, Packages, Interfaces
- Files, I/O streams, Java FX GUI

Exercise – 1:

- a) Develop a JAVA program to display default value of all primitive data type of JAVA
- b) Develop a java program that display the roots of a quadratic equation $ax^2+bx=0$. Find the discriminate D and basing on value of D, describe the nature of root.

Exercise – 2:

- a) Develop a JAVA program to search for an element in a given list of elements using binary search mechanism.
- b) Develop a JAVA program to sort for an element in a given list of elements using bubble sort
- c) Develop a JAVA program using String Buffer to delete, remove character.

Exercise – 3:

- a) Develop a JAVA program to implement class Mechanism. Build a class, methods and invoke them inside main method.
- b) Develop a JAVA program to implement method overloading.
- c) Develop a JAVA program to implement constructor.
- d) Develop a JAVA program to implement constructor overloading.

Exercise – 4:

- a) Develop a JAVA program to implement Single Inheritance.
- b) Develop a JAVA program to implement multi level Inheritance.
- c) Develop a JAVA program for abstract class to find areas of different shapes.

Exercise – 5:

- a) Develop a JAVA program on “super” keyword.
- b) Develop a JAVA program to implement Interface. What kind of Inheritance can be achieved?
- c) Develop a JAVA program that implements Runtime polymorphism.

Exercise – 6:

- a) Develop a JAVA program that describes exception handling mechanism.

- b) Develop a JAVA program on Multiple catch clauses.
- c) Develop a JAVA program to generate Built-in Exceptions.
- d) Develop a JAVA program to generate User Defined Exception.

Exercise – 7:

- a) Develop a JAVA program that creates threads by extending Thread class. First thread display “Good Morning “every 1 sec, the second thread displays “Hello “every 2 seconds and the third display “Welcome” every 3 seconds,(Repeat the same by implementing Runnable)
- b) Develop a program on isAlive() and join ().
- c) Demonstrate Daemon Threads.
- d) Demonstrate Producer Consumer Problem.

Exercise – 8:

- a) Develop a JAVA program that import and use the user defined packages.
- b) Demonstrate a GUI that display text in label and image in an ImageView (use JavaFX) Without writing any code.
- c) Develop a Tip Calculator app using several JavaFX components and learn how to respond to user interactions with the GUI

Textbooks:

- 1. Java one step ahead, Anitha Seth, B.L.Juneja, Oxford.
- 2. Joy with Java, Fundamentals of Object Oriented Programming, DebasisSamanta, MonalisaSarma, Cambridge, 2023.
- 3. Java 9 for Programmers, Paul Deitel, Harvey Deitel, 4th Edition, Pearson.

Reference Books:

- 1. The complete Reference Java, 11th edition, Herbert Schildt, TMH
- 2. Introduction to Java programming, 7th Edition, Y Daniel Liang, Pearson

Online Learning Resources:

- 1. <https://nptel.ac.in/courses/106/105/106105191/>
- 2. https://infyspringboard.onwingspan.com/web/en/app/toc/lex_auth_012880464547618816347_shared/overview

Semester	III	L	T	P	C	COURSE CODE
Regulation	V23	0	1	2	2	V23CSSE01
Name of the Course	Python Programming Lab					
Branch	Common to CSE, CST, CSE(AI) , AI&ML , CSE(DS) and EEE					

Syllabus Details

Course Outcomes: After Successful completion of the Course, the student will be able to:

CO1: Illustrate basic concepts and control structures in python Programming. **(K3)**

CO2: Demonstrate functions and packages. **(K3)**

CO3: Construct Python programs using structured data types. **(K3)**

CO4: Develop programs on Files, Exception handling and OOPs Concepts. **(K3)**

CO5: Construct programs for Data Analysis using Num Py and Pandas. **(K3)**

UNIT-I

History of Python Programming Language, Thrust Areas of Python, Installing Anaconda Python Distribution, Installing and Using Jupyter Notebook.

Parts of Python Programming Language: Identifiers, Keywords, Statements and Expressions, Variables, Operators, Precedence and Associativity, Data Types, Indentation, Comments, Reading Input, Print Output, Type Conversions, the type () Function and Is Operator, Dynamic and Strongly Typed Language.

Control Flow Statements: if statement, if-else statement, if...elif...else, Nested if statement, while Loop, for Loop, continue and break Statements, Catching Exceptions Using try and except Statement.

Sample Experiments:

1. Develop a program to find the largest element among three Numbers.
2. Develop a Program to display all prime numbers within an interval
3. Develop a program to swap two numbers without using a temporary variable.
4. Demonstrate the following Operators in Python with suitable examples.
i) Arithmetic Operators ii) Relational Operators iii) Assignment Operators iv) Logical Operators
v) Bit wise Operators vi) Ternary Operator vii) Membership Operators viii) Identity Operators
5. Develop a program to add and multiply complex numbers
6. Develop a program to print multiplication table of a given number.

UNIT-II

Functions: Built-In Functions, Commonly Used Modules, Function Definition and Calling the function, return Statement and void Function, Scope and Lifetime of Variables, Default Parameters, Keyword Arguments, *args and **kwargs, Command Line Arguments.

Strings: Creating and Storing Strings, Basic String Operations, Accessing Characters in String by Index Number, String Slicing and Joining, String Methods, Formatting Strings.

Lists: Creating Lists, Basic List Operations, Indexing and Slicing in Lists, Built-In Functions Used on Lists, List Methods, del Statement.

Sample Experiments:

1. Develop a program to define a function with multiple return values.
2. Develop a program to define a function using default arguments.
3. Develop a program to find the length of the string without using any library functions.

4. Develop a program to check if the substring is present in a given string or not.
5. Develop a program to perform the given operations on a list:
 - i. addition
 - ii. Insertion
 - iii. slicing
6. Develop a program to perform any 5 built-in functions by taking any list.

UNIT-III

Dictionaries: Creating Dictionary, Accessing and Modifying key: value Pairs in Dictionaries, Built-In Functions Used on Dictionaries, Dictionary Methods, del Statement.

Tuples and Sets: Creating Tuples, Basic Tuple Operations, tuple() Function, Indexing and Slicing in Tuples, Built-In Functions Used on Tuples, Relation between Tuples and Lists, Relation between Tuples and Dictionaries, Using zip() Function, Sets, Set Methods, Frozenset.

Sample Experiments:

1. Develop a program to create tuples (name, age, address, college) for at least two members and concatenate the tuples and print the concatenated tuples.
2. Develop a program to count the number of vowels in a string (No control flow allowed).
3. Develop a program to check if a given key exists in a dictionary or not.
4. Develop a program to add a new key-value pair to an existing dictionary.
5. Develop a program to sum all the items in a given dictionary.

UNIT-IV

Files: Types of Files, Creating and Reading Text Data, File Methods to Read and Write Data, Reading and Writing Binary Files, Pickle Module, Reading and Writing CSV Files, Python os and os.path Modules.

Object-Oriented Programming: Classes and Objects, Creating Classes in Python, Creating Objects in Python, Constructor Method, Classes with Multiple Objects, Class Attributes Vs Data Attributes, Encapsulation, Inheritance, Polymorphism.

Sample Experiments:

1. Develop a program to sort words in a file and put them in another file. The output file should have only lower-case words, so any upper-case words from source must be lowered.
2. Develop a Python program to print each line of a file in reverse order.
3. Develop a Python program to compute the number of characters, words and lines in a file.
4. Develop a program to create, display, append, insert and reverse the order of the items in the array.
5. Develop a program to add, transpose and multiply two matrices.
6. Develop a Python program to create a class that represents a shape. Include methods to calculate its area and perimeter. Implement subclasses for different shapes like circle, triangle, and square.

UNIT-V

Introduction to Data Science: Functional Programming, JSON and XML in Python, NumPy with Python, Pandas.

Sample Experiments:

1. Develop a Python program to check whether a JSON string contains complex object or not.
2. Demonstrate NumPy arrays creation using array () function.
3. Demonstrate use of ndim, shape, size, dtype.
4. Demonstrate basic slicing, integer and Boolean indexing.
5. Develop a Python program to find min, max, sum, cumulative sum of array
6. Prepare a dictionary with at least five keys and each key represent value as a list where this list contains at least ten values and convert this dictionary as a pandas data frame and explore the data through the data

frame as follows:

- a) Apply head () function to the pandas data frame
 - b) Perform various data selection operations on Data Frame
7. Select any two columns from the above data frame, and observe the change in one attribute with respect to other attribute with scatter and plot operations in matplotlib.

Reference Books:

1. Gowrishankar S, Veena A., Introduction to Python Programming, CRC Press.
2. Python Programming, S Sridhar, J Indumathi, V M Hariharan, 2ndEdition, Pearson, 2024
3. Introduction to Programming Using Python, Y. Daniel Liang, Pearson.

Online Learning Resources:

1. <https://www.coursera.org/learn/python-for-applied-data-science-ai>
2. <https://www.coursera.org/learn/python?specialization=python#syllabus>

Semester	IV	L	T	P	C	COURSE CODE
Regulation	V23	3	0	0	3	V23CST03
Name of the Course	Digital Logic & Computer Organization					
Branch	Common to CSE, CST, CSE(AI) , AI&ML and CSE(DS)					

Syllabus Details

Course Outcomes: After Successful completion of the Course, the student will be able to:

CO1: Explain different Data Representation and various Combinational Digital Logic Circuits.(K2)

CO2: Explain various Sequential Digital Logic Circuits and basic Structure of Computers.(K2)

CO3: Describe the basic concepts of Computer Arithmetic and Processor Organization.(K2)

CO4: Illustrate different types of Memory and organization of Memory.(K2)

CO5: Demonstrate the ways of accessing various Interfacing devices with Processor.(K3)

UNIT-I:

Data Representation: Binary Numbers, Fixed Point Representation. Floating Point Representation. Signed binary numbers.

Digital Logic Circuits I: Basic Logic Functions, Minimization of Logic expressions. K-Map Simplification, Combinational Circuits, Decoders, Multiplexers.

UNIT-II:

Digital Logic Circuits-II: Sequential Circuits, Flip-Flops, Binary counters, Registers, Shift Registers, Ripple counters

Basic Structure of Computers: Computer Types, Functional units, Basic operational concepts, Bus structures, Software, Performance, multiprocessors and multi-computers, Computer Generations, Von-Neumann Architecture

UNIT-III:

Computer Arithmetic: Addition and Subtraction of Signed Numbers, Design of Fast Adders, Multiplication of Positive Numbers, Signed-operand Multiplication, Fast Multiplication, Integer Division, Floating-Point Numbers and Operations

Processor Organization: Fundamental Concepts, Execution of a Complete Instruction, Multiple-Bus Organization, Hardwired Control and Multi programmed Control

UNIT-IV:

The Memory Organization: Basic Concepts, Semiconductor RAM Memories, Read-Only Memories, Speed, Size and Cost, Cache Memories, Performance Considerations, Virtual Memories, Memory Management Requirements, Secondary Storage

UNIT-V:

Input / Output Organization: Accessing I/O Devices, Interrupts, Processor Examples, Direct Memory Access, Buses, Interface Circuits, Standard I/O Interfaces

Textbooks:

1. Computer Organization, Carl Hamacher, Zvonko Vranesic, Safwat Zaky, 6th edition, McGraw-Hill
2. Digital Design, 6th Edition, M. Morris Mano, Pearson Education.

Reference Books:

1. Computer Organization and Architecture, William Stallings, 11th Edition, Pearson.
2. Computer Systems Architecture, M. Morris Mano, 3rd Edition, Pearson
3. Computer Organization and Design, David A. Paterson, John L. Hennessy, Elsevier
4. Fundamentals of Logic Design, Roth, 5th Edition, Thomson

Online Learning Resources:

1. <https://nptel.ac.in/courses/106/103/106103068/>

Semester	IV	L	T	P	C	COURSE CODE
Regulation	V23	3	0	0	3	V23CST07
Name of the Course	Database Management Systems					
Branch	Common to CSE, CST, CSE(AI) , AI&ML and CSE(DS)					

Syllabus Details

Course Outcomes: After Successful completion of the Course, the student will be able to:

CO1: Describe Database systems, various Data models and Database architecture. **(K2)**

CO2: Develop various real time applications using Relational algebra and Relational calculus. **(K3)**

CO3: Illustrate Nested queries and different types of joins. **(K3)**

CO4: Apply various Normalization techniques to refine schema. **(K3)**

CO5: Explain Transaction management, Concurrency control and Database indexing techniques. **(K2)**

UNIT-I

Introduction: Database system, Characteristics (Database Vs File System), Database Users, Advantages of Database systems, Database applications. Brief introduction of different Data Models; Concepts of Schema, Instance and data independence; Three tier schema architecture for data independence; Database system structure, environment, Centralized and Client Server architecture for the database.

Entity Relationship Model: Introduction, Representation of entities, attributes, entity set, relationship, relationship set, constraints, sub classes, super class, inheritance, specialization, generalization using ER Diagrams.

UNIT-II

Relational Model: Introduction to relational model, concepts of domain, attribute, tuple, relation, importance of null values, constraints (Domain, Key constraints, integrity constraints) and their importance, Relational Algebra, Relational Calculus.

BASIC SQL: Simple Database schema, data types, table definitions (create, alter), different DML operations (insert, delete, update).

UNIT-III

SQL: Basic SQL querying (select and project) using where clause, arithmetic & logical operations, SQL functions(Date and Time, Numeric, String conversion).Creating tables with relationship, implementation of key and integrity constraints, nested queries, sub queries, grouping, aggregation, ordering, implementation of different types of joins, view(updatable and non-updatable), relational set operations.

UNIT-IV

Schema Refinement (Normalization): Purpose of Normalization or schema refinement, concept of functional dependency, normal forms based on functional dependency Lossless join and dependency preserving decomposition, (1NF, 2NF and 3 NF), concept of surrogate key, Boyce-Codd normal form(BCNF), MVD, Fourth normal form(4NF), Fifth Normal Form (5NF).

UNIT-V

Transaction Concept: Transaction State, ACID properties, Concurrent Executions, Serializability, Recoverability, Implementation of Isolation, Testing for Serializability, lock based, time stamp based, optimistic, concurrency protocols, Deadlocks, Failure Classification, Storage, Recovery and Atomicity, Recovery algorithm.

Introduction to Indexing Techniques: B+ Trees, operations on B+Trees, Hash Based Indexing.

Textbooks:

1. Database Management Systems, 3rd edition, Raghurama Krishnan, Johannes Gehrke, TMH (For Chapters 2, 3, 4)
2. Database System Concepts, 5th edition, Silberschatz, Korth, Sudarsan, TMH (For Chapter 1 and Chapter 5)

Reference Books:

1. Introduction to Database Systems, 8th edition, C J Date, Pearson.
2. Database Management System, 6th edition, Ramez Elmasri, Shamkant B. Navathe, Pearson
3. Database Principles Fundamentals of Design Implementation and Management, Corlos Coronel, Steven Morris, Peter Robb, Cengage Learning.

Online Learning Resources:

1. <https://nptel.ac.in/courses/106/105/106105175/>
2. https://infyspringboard.onwingspan.com/web/en/app/toc/lex_auth_01275806667282022456_shared/overview.

Semester	IV	L	T	P	C	COURSE CODE
Regulation	V23	0	0	3	1.5	V23CSL07
Name of the Course	Database Management Systems Lab					
Branch	Common to CSE, CST, CSE(AI) , AI&ML and CSE(DS)					

Syllabus Details

Course Outcomes: After Successful completion of the Course, the student will be able to:

CO1: Construct SQL queries using DDL, DML and DCL commands. **(K3)**

CO2: Experiment with nested queries, built-in functions and various constraints. **(K3)**

CO3: Construct PL/SQL programs on Control structures, Cursors and Triggers . **(K3)**

CO4: Develop a Java program to connect with a database using JDBC. **(K3)**

Experiments covering the Topics:

- DDL, DML, DCL commands
- Queries, nested queries, built-in functions,
- PL/SQL programming- control structures
- Procedures, Functions, Cursors, Triggers,
- Database connectivity- ODBC/JDBC

Sample Programs:

1. Demonstrate Creation, altering and dropping of tables and inserting rows into a table (use constraints while creating tables) examples using SELECT command.
2. Construct queries (along with sub queries) using ANY, ALL, IN, EXISTS, NOTEXISTS, UNION, INTERSET, Constraints. Example:- Select the roll number and name of the student who secured fourth rank in the class.
3. Construct queries using Aggregate functions (COUNT, SUM, AVG, MAX and MIN), GROUP BY, HAVING and Creation and dropping of Views.
4. Construct queries using Conversion functions (to_char, to_number and to_date), string functions (Concatenation, lpad, rpad, ltrim, rtrim, lower, upper, initcap, length, substr and instr), date functions (Sysdate, next_day, add_months, last_day, months_between, least, greatest, trunc, round, to_char, to_date)
5.
 - i. Develop a simple PL/SQL program which includes declaration section, executable section and exception –Handling section (Ex. Student marks can be selected from the table and printed for those who secured first class and an exception can be raised if no records were found)
 - ii. Insert data into student table and use COMMIT, ROLLBACK and SAVEPOINT in PL/SQL block.
6. Develop a program that includes the features NESTED IF, CASE and CASE expression. The program can be extended using the NULLIF and COALESCE functions.
7. Develop programs using WHILE LOOPS, numeric FOR LOOPS, nested loops using ERROR Handling, BUILT –IN Exceptions, USE defined Exceptions, RAISE- APPLICATION ERROR.
8. Develop programs using procedures, passing parameters IN and OUT of PROCEDURES.
9. Develop programs on stored functions, invoke functions in SQL Statements and complex functions.
10. Develop programs using features parameters in a CURSOR, FOR UPDATE CURSOR, WHERE CURRENT of clause and CURSOR variables.
11. Develop Programs using BEFORE and AFTER Triggers, Row and Statement Triggers and INSTEAD OF Triggers
12. Construct a table and perform the search operation on table using indexing and non- indexing techniques.
13. Develop a Java program that connects to a database using JDBC
14. Develop a Java program to connect to a database using JDBC and insert values into it

15. Develop a Java program to connect to a database using JDBC and delete values from it

Textbooks:

1. Database Management Systems, 3rd edition, Raghurama Krishnan, Johannes Gehrke, TMH (For Chapters 2, 3, 4)
2. Database System Concepts, 5th edition, Silberschatz, Korth, Sudarsan, TMH (For Chapter 1 and Chapter 5)

Reference Books:

1. Oracle: The Complete Reference by Oracle Press
2. Nilesch Shah, "Database Systems Using Oracle", PHI, 2007
3. Rick F Vander Lans, "Introduction to SQL", Fourth Edition, Pearson Education, 2007

Semester	IV	L	T	P	C	COURSE CODE
Regulation	V23	0	1	2	2	V23CSSE02
Name of the Course	Full Stack Development – I					
Branch	Common to CSE, CST, CSE(AI) and AI&ML					

Syllabus Details

Course Outcomes: After Successful completion of the Course, the student will be able to:

CO1: Illustrate HTML elements and their attributes for designing static web pages. **(K3)**

CO2: Apply appropriate CSS styles to HTML elements. **(K3)**

CO3: Demonstrate JavaScript Pre-defined and User-defined Objects. **(K3)**

CO4: Develop dynamic web pages and validate forms using JavaScript. **(K3)**

Experiments covering the Topics:

- Lists, Links and Images
- HTML Tables, Forms and Frames
- HTML 5 and Cascading Style Sheets, Types of CSS
- Selector forms
- CSS with Color, Background, Font, Text and CSS Box Model
- Applying JavaScript - internal and external, I/O, Type Conversion
- JavaScript Conditional Statements and Loops, Pre-defined and User-defined Objects
- JavaScript Functions and Events
- Node.js

Sample Programs:

1. Lists, Links and Images

- a) Develop a HTML program, to explain the working of lists.
- b) Note: It should have an ordered list, unordered list, nested lists and ordered list in an unordered list and definition lists.
- c) Develop a HTML program, to explain the working of hyperlinks using <a> tag and href, target Attributes.
- d) Develop a HTML document that has your image and your friend's image with a specific height and width. Also when clicked on the images it should navigate to their respective profiles.
- e) Develop a HTML program, in such a way that, rather than placing large images on a page, the preferred technique is to use thumbnails by setting the height and width parameters to something like to 100*100 pixels. Each thumbnail image is also a link to a full sized version of the image. Create an image gallery using this technique

2. HTML Tables, Forms and Frames

- a) Develop a HTML program, to explain the working of tables. (use tags: <table>, <tr>, <th>, <td> and attributes: border, rowspan, colspan)
- b) Develop a HTML program, to explain the working of tables by preparing a timetable. (Note: Use <caption> tag to set the caption to the table & also use cell spacing, cell padding, border, rowspan, colspan etc.).
- c) Develop a HTML program, to explain the working of forms by designing Registration form. (Note: Include text field, password field, number field, date of birth field, checkboxes, radio buttons, list boxes using <select> & <option> tags, <text area> and two buttons ie: submit and reset. Use tables to provide a

better view).

- d) Develop a HTML program, to explain the working of frames, such that page is to be divided into 3 parts on either direction. (Note: first frame □ image, second frame □ paragraph, third frame □ hyperlink. And also make sure of using “no frame” attribute such that frames to be fixed).

3. HTML 5 and Cascading Style Sheets, Types of CSS

- a) Develop a HTML program, that makes use of <article>, <aside>, <figure>, <figcaption>,<footer>, <header>, <main>, <nav>, <section>, <div>, tags.
b) Develop a HTML program, to embed audio and video into HTML web page.
c) Develop a program to apply different types (or levels of styles or style specification formats)- inline, internal, external styles to HTML elements. (identify selector, property and value).

4. Selector forms

- a) Develop a program to apply different types of selector forms
i. Simple selector (element, id, class, group, universal)
ii. Combinator selector (descendant, child, adjacent sibling, general sibling)
iii. Pseudo-class selector
iv. Pseudo-element selector
v. Attribute selector

5. CSS with Color, Background, Font, Text and CSS Box Model

- a) Develop a program to demonstrate the various ways you can reference a color in CSS.
b) Develop a CSS rule that places a background image halfway down the page, tilting it horizontally. The image should remain in place when the user scrolls up or down.
c) Develop a program using the following terms related to CSS font and text:
i. font-size ii. font-weight iii. font-style
iv. text-decoration v. text-transformation vi. text-alignment
d) Develop a program, to explain the importance of CSS Box model using
i. Content ii. Border iii. Margin iv. padding

6. Applying JavaScript - internal and external, I/O, Type Conversion

- a) Develop a program to embed internal and external JavaScript in a web page.
b) Develop a program to explain the different ways for displaying output.
c) Develop a program to explain the different ways for taking input.
d) Create a webpage which uses prompt dialogue box to ask a voter for his name and age. Display the information in table format along with either the voter can vote or not

7. Java Script Pre-defined and User-defined Objects

- a) Develop a program using document object properties and methods.
b) Develop a program using window object properties and methods.
c) Develop a program using array object properties and methods.
d) Develop a program using math object properties and methods.
e) Develop a program using string object properties and methods.
f) Develop a program using regex object properties and methods.
g) Develop a program using date object properties and methods.
h) Develop a program to explain user-defined object by using properties, methods, accessors, constructors and display.

8. Java Script Conditional Statements and Loops

- a) Develop a program which asks the user to enter three integers, obtains the numbers from the user and outputs HTML text that displays the larger number followed by the words “LARGER NUMBER” in an

- information message dialog. If the numbers are equal, output HTML text as “EQUAL NUMBERS”.
- b) Develop a program to display week days using switch case.
 - c) Develop a program to print 1 to 10 numbers using for, while and do-while loops.
 - d) Develop a program to print data in object using for-in, for-each and for-of loops
 - e) Develop a program to determine whether a given number is an ‘ARMSTRONG NUMBER’ or not.
[Eg: 153 is an Armstrong number, since sum of the cube of the digits is equal to the number i.e., $1^3 + 5^3 + 3^3 = 153$]
 - f) Develop a program to display the denomination of the amount deposited in the bank in terms of 100’s, 50’s, 20’s, 10’s, 5’s, 2’s & 1’s. (Eg: If deposited amount is Rs.163, the output should be 1-100’s, 1-50’s, 1- 10’s, 1-2’s & 1-1’s)

9. Java Script Functions and Events

- a) Design a appropriate function should be called to display
 - i. Factorial of that number
 - ii. Fibonacci series up to that number
 - iii. Prime numbers up to that number
 - iv. Is it palindrome or not
- b) Prepare a HTML file having a text box and four buttons named Factorial, Fibonacci, Prime, and Palindrome. When a button is pressed an appropriate function should be called to display
 - i. Factorial of that number
 - ii. Fibonacci series up to that number
 - iii. Prime numbers up to that number
 - iv. Is it palindrome or not
- c) Develop a program to validate the following fields in a registration page
 - i. Name (start with alphabet and followed by alphanumeric and the length should not be less than 6 characters)
 - ii. Mobile (only numbers and length 10 digits)
 - iii. E-mail (should contain format like xxxxxxxx@xxxxxx.xxx)

Reference Books:

1. Programming the World Wide Web, 7th Edition, Robert W Sebesta, Pearson, 2013.
2. Web Programming with HTML5, CSS and JavaScript, John Dean, Jones & Bartlett Learning, 2019 (Chapters 1-11).
3. Pro MERN Stack: Full Stack Web App Development with Mongo, Express, React, and Node, Vasan Subramanian, 2nd edition, APress, O’Reilly.

Online Learning Resources:

1. <https://www.w3schools.com/html>
2. <https://www.w3schools.com/css>
3. <https://www.w3schools.com/js/>
4. <https://www.w3schools.com/nodejs>